Chapter 13 Discrete Distributions - Homework Answers

Chapter revision | Warm-up

1 **a** There are 12 court cards, so $P(\text{court card}) = \frac{12}{52} = \frac{3}{13}$.

b $P(\text{not a court card}) = 1 - \frac{3}{13} = \frac{10}{13}$.

c There are 13 hearts and 4 kings, and the king of hearts is counted in both. So $P(\text{heart or king}) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.

2 **a** The tosses are independent, so $P(HHH) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.

b $P(\text{at least one head}) = 1 - P(TTT) = 1 - \frac{1}{8} = \frac{7}{8}$.

c The three outcomes with two heads are HHT, HTH and THH, each with probability $\frac{1}{8}$. So $P(\text{exactly two heads}) = \frac{3}{8}$.

3 **a** The stages are independent: $P(RR) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$.

b With no replacement, after one red is removed there are 2 red in 4: $P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$.

4 **a** $f_r = \frac{f}{50}$: $0.12,\ 0.28,\ 0.36,\ 0.24$. Check: $0.12 + 0.28 + 0.36 + 0.24 = 1$.

b $\bar{x} = \sum x\, f_r = 1(0.12) + 2(0.28) + 3(0.36) + 4(0.24) = 0.12 + 0.56 + 1.08 + 0.96 = 2.72$

c $\sum x^2 f_r = 1(0.12) + 4(0.28) + 9(0.36) + 16(0.24) = 0.12 + 1.12 + 3.24 + 3.84 = 8.32$, so $\sigma^2 = 8.32 - 2.72^2 = 8.32 - 7.3984 = 0.9216$ and $\sigma = \sqrt{0.9216} = 0.96$.

5 **a** $3 + 6 + 9 + 12 = \sum_{k=1}^{4} 3k = 3(1 + 2 + 3 + 4) = 30$

b $\sum (3x + 1)p(x) = 3\sum x\,p(x) + \sum p(x) = 3(2) + 1 = 7$

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13A | The language of probability distributions

Lesson Examples

C1 **Tutor demo.** Each die shows a six with probability $\frac{1}{6}$, and the 36 outcomes for the two dice are equally likely. No sixes: $5 \times 5 = 25$ outcomes. Two sixes: 1 outcome. One six: $36 - 25 - 1 = 10$ outcomes.

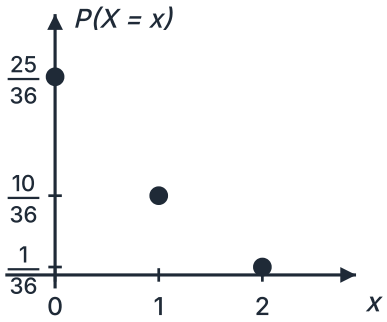
a

x	0	1	2
$P(X = x)$	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$

b $P(X \geq 1) = \frac{10}{36} + \frac{1}{36} = \frac{11}{36}$

$P(X \text{ is even}) = \frac{25}{36} + \frac{1}{36} = \frac{13}{18}$

c



Your turn. Each throw is less than 3 with probability $\frac{2}{6} = \frac{1}{3}$, and more than 2 with probability $\frac{2}{3}$. So $P(X = 0) = \frac{2}{3} \times \frac{2}{3}$, $P(X = 2) = \frac{1}{3} \times \frac{1}{3}$, and $P(X = 1) = \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{3}$.

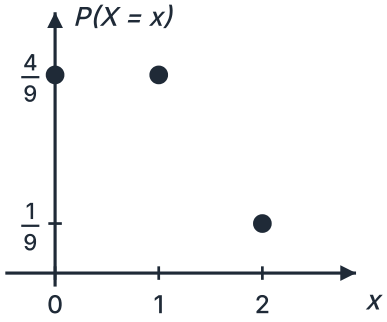
a

x	0	1	2
$P(X = x)$	$\frac{4}{9}$	$\frac{4}{9}$	$\frac{1}{9}$

b $P(X \leq 1) = \frac{4}{9} + \frac{4}{9} = \frac{8}{9}$

$P(X \text{ is odd}) = P(X = 1) = \frac{4}{9}$

c



C2 **Tutor demo.** Divide each number by the total 1200.

a

x	Red	Blue	Gold	Green
$P(X = x)$	0.26	0.24	0.2	0.3

b $P(X = \text{Red or Gold}) = 0.26 + 0.2 = 0.46$

c Categorical: the values are houses, not numbers.

Your turn. Divide each number by the total 50.

a

x	red	green	yellow	black
$P(X = x)$	0.36	0.24	0.3	0.1

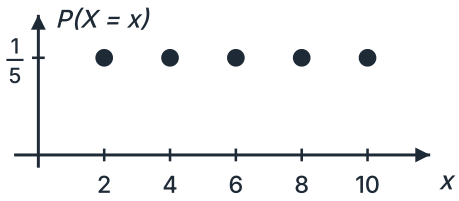
b $P(X = \text{green or yellow}) = 0.24 + 0.3 = 0.54$

c Categorical: the values are colours, not numbers.

C3 **Tutor demo.** Each card has probability $\frac{1}{5}$.

x	2	4	6	8	10
$P(X = x)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

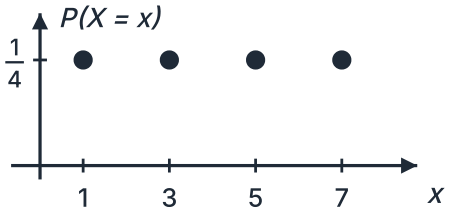
The distribution is uniform, because all five values have the same probability.



Your turn. Each face has probability $\frac{1}{4}$.

x	1	3	5	7
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

The distribution is uniform, because all four values have the same probability.



C4 **Tutor demo.** Write S for a six and N for not a six. The throws are independent.

a $P(X = 1) = P(\text{S}) = \frac{1}{6}$, $P(X = 2) = P(\text{NS}) = \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$, $P(X = 3) = P(\text{NNS}) = \frac{25}{216}$, $P(X = 4) = P(\text{NNNS}) = \frac{125}{1296}$

b

x	1	2	3	4	$\cdots$	n	$\cdots$
$P(X = x)$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{25}{216}$	$\frac{125}{1296}$	$\cdots$	$\frac{5^{n-1}}{6^n}$	$\cdots$

c The values are the whole numbers $1, 2, 3, 4, \ldots$ The list never ends, but every possible number of throws appears in it once and once only, so the set is countably infinite.

Your turn. Write R for red and B for blue, with $P(\text{R}) = \frac{1}{3}$ and $P(\text{B}) = \frac{2}{3}$.

a $P(X = 1) = \frac{1}{3}$, $P(X = 2) = P(\text{BR}) = \frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$, $P(X = 3) = P(\text{BBR}) = \frac{4}{27}$, $P(X = 4) = P(\text{BBBR}) = \frac{8}{81}$

b

x	1	2	3	4	$\cdots$	n	$\cdots$
$P(X = x)$	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{4}{27}$	$\frac{8}{81}$	$\cdots$	$\frac{2^{n-1}}{3^n}$	$\cdots$

c The values are the whole numbers $1, 2, 3, 4, \ldots$, which can be listed in order, even though the list never ends.

C5 **Tutor demo.**

a Numeric, continuous

b Numeric, discrete

c Categorical

d Numeric, discrete (the recorded times go up in steps of 0.01 seconds, so they can be listed)

Your turn.

a Numeric, continuous

b Numeric, discrete

c Categorical

d Numeric, discrete (the recorded masses are whole numbers of grams, so they can be listed)

Textbook Homework

1

a Numeric, discrete
b Numeric, continuous
c Categorical
d Numeric, continuous
e Categorical
f Categorical
g On a standard scale of shoe sizes, this is numeric and discrete. The length of a person's foot would be a numeric, continuous distribution.
h Numeric, discrete. Reported ATAR scores are between 30 and 99.95 in steps of 0.05. There are around 1400 different scores awarded.

2

a

outcome	HH	HT	TH	TT
probability	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Uniform distribution (and categorical).

b

outcome	2 heads	1 head and 1 tail	2 tails
probability	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

3

a

outcome	red	green
probability	$\frac{4}{7}$	$\frac{3}{7}$

b

outcome	Jack	Kylie	Lochlan	Other
probability	0.06	0.08	0.04	0.82

c

outcome	P	A	R	M	T
probability	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{5}$

d

outcome	1	2	3	4
probability	$\frac{9}{1000}$	$\frac{90}{1000}$	$\frac{900}{1000}$	$\frac{1}{1000}$

e

outcome	even	prime	neither
probability	$\frac{5}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

4

a Let X be the number of letters in a randomly chosen word.

x	3	4	6
$P(X=x)$	$\frac{5}{8}$	$\frac{1}{4}$	$\frac{1}{8}$

b Let X be the number of heads recorded when 2 coins are thrown.

x	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

c Let X be the digit chosen from the 12 digits of 1.41421356237.

x	1	2	3	4	5	6	7
$P(X=x)$	$\frac{3}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

d Let X be the number selected.

x	1	2	3	4	5
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

(The answer is the same if the two bags are combined into one bag of eight marbles, because the two bags hold the same number of marbles.)

5 **a** {T}, {F1}, {F2}, {T, F1}, {T, F2}, {F1, F2}, {T, F1, F2}

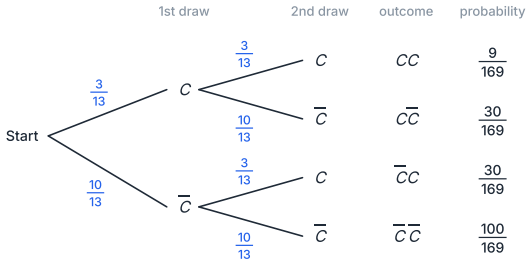
b

x	5	10	15	20
$P(X=x)$	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{1}{7}$

6 **a** Yes **b** No **c** Yes **d** Yes **e** No **f** Yes

7 **a** 0.2 **b** 0.6 **c** 0.75 **d** 0 **e** 0.6 **f** 0.85 **g** 0.9 **h** 0.7 **i** 0.45

8 **a** (i) Let C be the event 'A court card is drawn', with $P(C) = \frac{12}{52} = \frac{3}{13}$.



(ii)

x	0	1	2
$P(X=x)$	$\frac{100}{169}$	$\frac{60}{169}$	$\frac{9}{169}$

b (i) The eight outcomes EEE, EEO, EOE, EOO, OEE, OEO, OOE, OOO each have probability $\frac{1}{8}$.

(ii)

x	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

c GGG has probability $\frac{8}{125}$; GGB, GBG, BGG each have probability $\frac{12}{125}$; GBB, BGB, BBG each have probability $\frac{18}{125}$; BBB has probability $\frac{27}{125}$. Let X be the number of girls called.

x	0	1	2	3
$P(X=x)$	$\frac{27}{125}$	$\frac{54}{125}$	$\frac{36}{125}$	$\frac{8}{125}$

d Let S be the event 'A wallaby from Snake Gully was selected', with $P(S) = \frac{6}{20} = 0.3$. SSS has probability 0.027; $\overline{S}SS$, $S\overline{S}S$, $SS\overline{S}$ each have probability 0.063; $\overline{S}\,\overline{S}S$, $\overline{S}S\overline{S}$, $S\overline{S}\,\overline{S}$ each have probability 0.147; $\overline{S}\,\overline{S}\,\overline{S}$ has probability 0.343.

x	0	1	2	3
$P(X=x)$	0.343	0.441	0.189	0.027

9 **a** $a = \frac{1}{25}$ **b** $a = \frac{1}{14}$ **c** $a = \frac{1}{27}$ **d** $a = \frac{1}{10}$ **e** $a = 1$

10 **a** (i) EE and OO each have probability $\frac{1}{5}$; EO and OE each have probability $\frac{3}{10}$.

(ii)

x	0	1	2
$P(X=x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$

b BB has probability $\frac{2}{5}$; BG and GB each have probability $\frac{4}{15}$; GG has probability $\frac{1}{15}$.

x	0	1	2
$P(X=x)$	$\frac{2}{5}$	$\frac{8}{15}$	$\frac{1}{15}$

c (i) EE has probability $\frac{3}{10}$; ER, RE, ET, TE each have probability $\frac{3}{20}$; RT and TR each have probability $\frac{1}{20}$.

(ii)

x	0	1	2
$P(X=x)$	$\frac{1}{10}$	$\frac{3}{5}$	$\frac{3}{10}$

11

x	22	44	55	24 or 42	25 or 52	45 or 54
$P(X=x)$	$\frac{1}{9}$	$\frac{1}{4}$	$\frac{1}{36}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{6}$

12

a

outcome	RR	RG	GR	GG
probability	$\frac{16}{49}$	$\frac{12}{49}$	$\frac{12}{49}$	$\frac{9}{49}$

b

outcome	RR	RG	GR	GG
probability	$\frac{12}{42}$	$\frac{12}{42}$	$\frac{12}{42}$	$\frac{6}{42}$

c (H, D, S, C stand for hearts, diamonds, spades, clubs.)

outcome	HH	DD	SS	CC
probability	$\frac{1}{17}$	$\frac{1}{17}$	$\frac{1}{17}$	$\frac{1}{17}$

outcome	HS or SH	HC or CH	HD or DH	SC or CS	SD or DS	CD or DC
probability	$\frac{13}{102}$	$\frac{13}{102}$	$\frac{13}{102}$	$\frac{13}{102}$	$\frac{13}{102}$	$\frac{13}{102}$

13

d There is no guarantee that the results will be identical, though you would expect more *trials* (repeats of the experiment) to bring your results closer to each other and to the theoretical probabilities.
e Theoretical results: $P(X=0)=0.3$, $P(X=1)=0.6$, $P(X=2)=0.1$
f It might be easier to perform the experiment with coloured balls or tokens. Running the experiment in pairs with a nominated recorder also helps. The paper pieces need to be indistinguishable and well mixed in the bag. You could increase the number of trials or combine the class results.

14

EEE and OOO each have probability $\frac{1}{20}$; the other six possible outcomes each have probability $\frac{3}{20}$.

x	0	1	2	3
$P(X=x)$	$\frac{1}{20}$	$\frac{9}{20}$	$\frac{9}{20}$	$\frac{1}{20}$

15

a The condition that the sum of the probabilities is 1 gives $a=\frac{1}{4}$ or $a=1$. But $a=1$ gives probabilities outside the interval $0\le p\le 1$, and the only valid answer is $a=\frac{1}{4}$.
b $a=1$ or $a=\frac{7}{6}$ (both are valid)

16

a Let X be the sum of the numbers on the three cards. This question is best done by asking what card is *discarded*.

x	20	21	22
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

b

x	20	21	22
$P(X=x)$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{3}{10}$

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13A | The language of probability distributions

Past-Paper Homework - Worked Solutions

PP 1 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)(i)**

The two dice give $6 \times 6 = 36$ equally likely outcomes. The outcomes with a sum of 7 are $(1,6)$, $(2,5)$, $(3,4)$, $(4,3)$, $(5,2)$ and $(6,1)$, which is 6 outcomes. So

$$p = P(\text{sum of } 7) = \frac{6}{36} = \frac{1}{6}.$$

The result is a categorical distribution with two values: '7' with probability $\frac{1}{6}$ and 'not a 7' with probability $\frac{5}{6}$.

PP 2 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)(i)(iv)**

(i) X is the number of successes in 4 independent trials, each with probability p of success. The value $X = 4$ means that all four trials succeed, so by the product rule

$$P(X = 4) = p \times p \times p \times p = p^4.$$

The table gives $P(X = 4) = 0.0081$, so $p^4 = 0.0081 = 0.3^4$. Since p is a probability, $p \geq 0$, and so $p = 0.3$.

(Check: $X = 0$ means all four trials fail, and $(1 - 0.3)^4 = 0.7^4 = 0.2401$, which is the first entry of the table.)

(iv) The event $X \geq 3$ happens when $X = 3$ or $X = 4$. These are different values, so the probabilities add. Both are given in the table:

$$P(X \geq 3) = P(X = 3) + P(X = 4) = 0.0756 + 0.0081 = 0.0837.$$

PP 3 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(ii)**

The complement of 'at least one incorrect answer' is 'all five answers correct'. Each answer is correct with probability $\frac{1}{4}$, and the five choices are independent, so

$$P(\text{all correct}) = \left(\frac{1}{4}\right)^5 = \frac{1}{1024}.$$

Hence

$$P(\text{at least one incorrect}) = 1 - \frac{1}{1024} = \frac{1023}{1024}.$$

PP 4 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q11(a)(iii)**

Each shot is a bullseye with probability $\frac{1}{3}$, so it misses with probability $\frac{2}{3}$. Suppose the archer uses n shots, and the shots are independent. By the product rule,

$$P(\text{no bullseye in } n \text{ shots}) = \left(\frac{2}{3}\right)^n, \quad \text{so} \quad P(\text{at least one bullseye}) = 1 - \left(\frac{2}{3}\right)^n.$$

For 90% certainty we need

$$1 - \left(\frac{2}{3}\right)^n \geq 0.9, \quad \text{that is,} \quad \left(\frac{2}{3}\right)^n \leq 0.1.$$

Take logarithms. Since $\ln \frac{2}{3} < 0$, the inequality sign reverses when we divide by it:

$$n \ln \tfrac{2}{3} \leq \ln 0.1, \qquad n \geq \frac{\ln 0.1}{\ln \frac{2}{3}} = 5.678\ldots$$

So the archer should use 6 shots. (Check: $\left(\frac{2}{3}\right)^5 = \frac{32}{243} \approx 0.132$, which is more than 0.1, but $\left(\frac{2}{3}\right)^6 = \frac{64}{729} \approx 0.088$, which is less than 0.1.)

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13B | Expected value

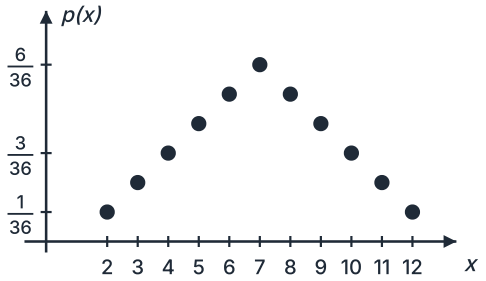
Lesson Examples

C1 **Tutor demo.** The 36 outcomes for the two dice are equally likely. The sum 7 comes from 6 of them: $(1,6)$, $(2,5)$, $(3,4)$, $(4,3)$, $(5,2)$, $(6,1)$. Each step away from 7 loses one outcome, down to one outcome each for the sums 2 and 12.

a

x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

b



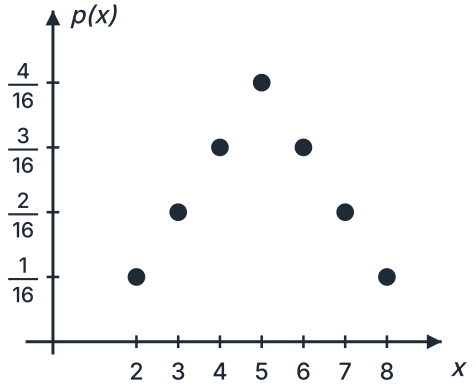
c The graph is symmetric about $x = 7$: each value $7 - k$ on the left has the same probability as the value $7 + k$ on the right. So the centre of the distribution is $x = 7$.

Your turn. The $4 \times 4 = 16$ outcomes are equally likely. The sum 5 comes from 4 of them: $(1,4)$, $(2,3)$, $(3,2)$, $(4,1)$.

a

x	2	3	4	5	6	7	8
$p(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

b

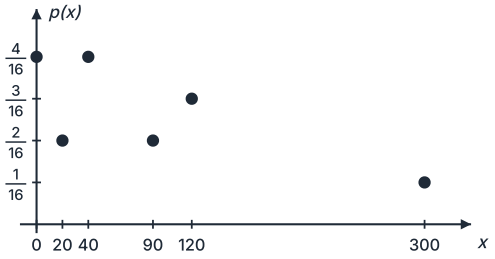


c The graph is symmetric about $x = 5$, so the centre of the distribution is $x = 5$.

C2 **Tutor demo.** Let $\$x$ be the amount the pickpocket takes. Each friend is chosen with probability $\frac{1}{16}$.

x	0	20	40	90	120	300	Sum
$p(x)$	$\frac{4}{16}$	$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{1}{16}$	1
$xp(x)$	0	2.5	10	11.25	22.5	18.75	65

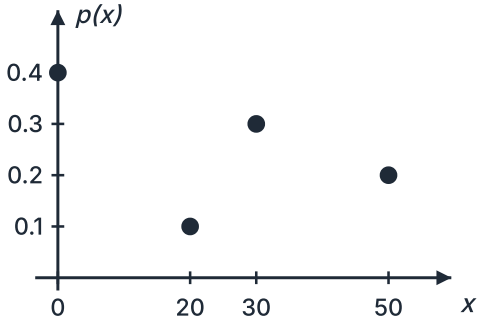
Hence the pickpocket's expected gain is $\$65$.



Your turn. Let $\$x$ be the amount the thief takes. Each friend is chosen with probability $\frac{1}{10}$.

x	$p(x)$	$xp(x)$
0	0.4	0
20	0.1	2
30	0.3	9
50	0.2	10
Sum	1	21

Hence the thief's expected gain is $\$21$.



C3 **Tutor demo.** Each value has probability $\frac{1}{8}$.

x	1	2	3	4	5	6	7	8	Sum
$p(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	1
$xp(x)$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$	$\frac{4}{8}$	$\frac{5}{8}$	$\frac{6}{8}$	$\frac{7}{8}$	$\frac{8}{8}$	$\frac{36}{8}$

Hence $\mu = \frac{36}{8} = 4\frac{1}{2}$. This is not one of the numbers on the die.



Your turn. Each value has probability $\frac{1}{12}$. The $xp(x)$ row is $\frac{1}{12}, \frac{2}{12}, \ldots, \frac{12}{12}$, with sum $\mathrm{E}(X) = \frac{1+2+\cdots+12}{12} = \frac{78}{12} = 6\frac{1}{2}$.

So $\mu = 6\frac{1}{2}$. This is not one of the numbers on the die.



Textbook Homework

1

a

x	0	1	2	3	Sum
$p(x)$	0.4	0.1	0.2	0.3	1
$xp(x)$	0	0.1	0.4	0.9	1.4

Hence $\mathrm{E}(X) = 1.4$.

b

x	2	4	6	8	Sum
$p(x)$	0.1	0.4	0.4	0.1	1
$xp(x)$	0.2	1.6	2.4	0.8	5

Hence $\mathrm{E}(X) = 5$.

c

x	-50	-20	0	30	100	Sum
$p(x)$	0.1	0.35	0.4	0.1	0.05	1
$xp(x)$	-5	-7	0	3	5	-4

Hence $\mathrm{E}(X) = -4$.

2

a

x	$p(x)$	$xp(x)$
-40	$\frac{1}{2}$	-20
0	$\frac{1}{6}$	0
30	$\frac{1}{6}$	5
60	$\frac{1}{6}$	10
Sum	1	-5

b Expected value $= -5$

c The average cost to the player per game is 5 cents.

d $100 \times (-5) = -500$ cents. Thus the player expects to lose 500 cents and the casino expects to make 500 cents profit. This is an expected average value, not guaranteed.

4

a

x_i	2	4	6	8	10	Sum
p_i	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	1
x_ip_i	$\frac{2}{5}$	$\frac{4}{5}$	$\frac{6}{5}$	$\frac{8}{5}$	$\frac{10}{5}$	6

So $\mathrm{E}(X) = 6$.

b

x_i	-3	1	2	5	6	Sum
p_i	0.1	0.3	0.2	0.3	0.1	1
x_ip_i	-0.3	0.3	0.4	1.5	0.6	2.5

So $\mathrm{E}(X) = 2.5$.

5

a

x	1.50	2.10	2.40	Sum
$p(x)$	$\frac{5}{12}$	$\frac{4}{12}$	$\frac{3}{12}$	1
$xp(x)$	0.625	0.7	0.60	1.925

The expected value is $\$1.925$.

b If 100 purchases are made at random the expected cost is $\$192.50$.

6

a $\mathrm{E}(X) = 3$

b (i) $\mathrm{E}(Y) = 6$ (ii) Yes

c (i) $\mathrm{E}(Z) = 4$ (ii) Yes

7

a 15 **b** 10 **c** $\frac{5}{2}$ **d** 3 **e** 0 **f** 18

8

x	0	1	2	3	Sum
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1
$xp(x)$	0	$\frac{3}{8}$	$\frac{6}{8}$	$\frac{3}{8}$	$\frac{12}{8}$

The expected value is $1\frac{1}{2}$, as might be expected from the symmetry of the table of probabilities.

9

x	0	1	2	Sum
$p(x)$	$\frac{19}{34}$	$\frac{13}{34}$	$\frac{1}{17}$	1
$xp(x)$	0	$\frac{13}{34}$	$\frac{2}{17}$	$\frac{17}{34}$

The expected value is $\frac{1}{2}$.

10

d

x	0	1	2	3	4	5	Sum
$p(x)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1
$xp(x)$	0	$\frac{10}{36}$	$\frac{16}{36}$	$\frac{18}{36}$	$\frac{16}{36}$	$\frac{10}{36}$	$\frac{70}{36}$

Hence $\mathrm{E}(X) = \frac{35}{18}$.

f In any dice experiment, it is important to check the randomness of your dice rolls. This can depend on your rolling technique. Try throwing a die 12 times and see if every outcome is equally likely. Does each outcome seem independent of the last?

11

a $\frac{3}{15}, \frac{3}{15}, \frac{3}{15}, \frac{2}{15}, \frac{2}{15}, \frac{2}{15}$

b -12, so the bank expects to make 12 cents each game, on average.

12

a $P(\text{Orange}) = \frac{1}{6}$, $P(\text{Strawberry}) = \frac{2}{6}$, $P(\text{Apple}) = \frac{3}{6}$

b

outcome	OOO	SSS	AAA	Other	Sum
x	$11k$	$2k$	k	0	—
$p(x)$	$\frac{1}{216}$	$\frac{8}{216}$	$\frac{27}{216}$	$\frac{180}{216}$	1

c Breaking even means $\mathrm{E}(X) = 1$, the \$1 paid in: $\mathrm{E}(X) = \frac{11k \times 1 + 2k \times 8 + k \times 27}{216} = \frac{54k}{216} = \frac{k}{4} = 1$, so $k = 4$ and OOO pays $11k = 44$ dollars. The payout will be \$44 and their profit would be \$43, accounting for the \$1 entry fee.

13

b

$$\mu = 1 \times \tfrac{1}{2} + 2 \times \tfrac{1}{4} + 3 \times \tfrac{1}{8} + 4 \times \tfrac{1}{16} + 5 \times \tfrac{1}{32} + 6 \times \tfrac{1}{64} + \cdots \quad (1)$$

$$\text{Doubling,} \quad 2\mu = 1 \times 1 + 2 \times \tfrac{1}{2} + 3 \times \tfrac{1}{4} + 4 \times \tfrac{1}{8} + 5 \times \tfrac{1}{16} + 6 \times \tfrac{1}{32} + \cdots \quad (2)$$

$$\text{Subtracting (1) from (2),} \quad \mu = 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \tfrac{1}{32} + \cdots \quad (3)$$

$$\text{Doubling,} \quad 2\mu = 2 + 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \cdots \quad (4)$$

Subtracting (3) from (4), $\mu = 2$.

c On average, we would expect to get a head on the second throw. You could test this by recording how many throws it takes over say 50 trials and averaging the results.

14

$$\begin{aligned}\mathrm{E}(X) &= 2 \times \tfrac{1}{2} + 4 \times \tfrac{1}{4} + 8 \times \tfrac{1}{8} + 16 \times \tfrac{1}{16} + \cdots \\ &= 1 + 1 + 1 + 1 + \cdots\end{aligned}$$

The expected value 'increases without bound', that is, $\mathrm{E}(X) \to \infty$ as the game continues. This suggests that there is no reasonable price the casino could put on this game and expect to break even. There are various issues with this scenario in real life. Casinos would not provide a game which had no upper limit to the payout. Patrons would also be unwilling to pay a large price for a game with such low apparent probabilities for the later stages of the game. The calculation of a simple expected value may not be the best way to analyse this game.

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13C | Variance and standard deviation

Lesson Examples

C1 **Tutor demo.** Each value has probability $\frac{1}{8}$.

x	1	2	3	4	5	6	7	8	Sum
$p(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	1
$xp(x)$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$	$\frac{4}{8}$	$\frac{5}{8}$	$\frac{6}{8}$	$\frac{7}{8}$	$\frac{8}{8}$	$4\frac{1}{2}$
$(x-\mu)^2$	$\frac{49}{4}$	$\frac{25}{4}$	$\frac{9}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{9}{4}$	$\frac{25}{4}$	$\frac{49}{4}$	—
$(x-\mu)^2p(x)$	$\frac{49}{32}$	$\frac{25}{32}$	$\frac{9}{32}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{9}{32}$	$\frac{25}{32}$	$\frac{49}{32}$	$\frac{168}{32}$

Hence the mean is $4\frac{1}{2}$ and the variance is $\frac{168}{32} = 5\frac{1}{4}$.

Your turn. Each value has probability $\frac{1}{4}$.

x	1	2	3	4	Sum
$p(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	1
$xp(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$2\frac{1}{2}$
$(x-\mu)^2$	$\frac{9}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{9}{4}$	—
$(x-\mu)^2p(x)$	$\frac{9}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{9}{16}$	$\frac{20}{16}$

Hence the mean is $2\frac{1}{2}$ and the variance is $\frac{20}{16} = 1\frac{1}{4}$.

C2 **Tutor demo.** The values are 1, 2 and 3, on one, two and three of the six faces.

x	1	2	3	Sum
$p(x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	1
$xp(x)$	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{9}{6}$	$\frac{14}{6}$
$x^2p(x)$	$\frac{1}{6}$	$\frac{8}{6}$	$\frac{27}{6}$	$\frac{36}{6}$

From the third row, $\mathrm{E}(X) = \frac{14}{6} = 2\frac{1}{3}$. From the last row, $\mathrm{E}(X^2) = \frac{36}{6} = 6$, so

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 6 - \left(\tfrac{7}{3}\right)^2 = 6 - \tfrac{49}{9} = \tfrac{5}{9}.$$

Your turn. The values are 1, 2 and 4, on two, three and one of the six faces.

x	1	2	4	Sum
$p(x)$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{1}{6}$	1
$xp(x)$	$\frac{2}{6}$	$\frac{6}{6}$	$\frac{4}{6}$	$\frac{12}{6}$
$x^2p(x)$	$\frac{2}{6}$	$\frac{12}{6}$	$\frac{16}{6}$	$\frac{30}{6}$

From the third row, $\mathrm{E}(X) = \frac{12}{6} = 2$. From the last row, $\mathrm{E}(X^2) = \frac{30}{6} = 5$, so $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 5 - 2^2 = 1$.

C3 **Tutor demo.** Let $\$x$ be the amount the pickpocket takes. Each friend is chosen with probability $\frac{1}{16}$.

x	0	20	40	90	120	300	Sum
$p(x)$	$\frac{4}{16}$	$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{1}{16}$	1
$xp(x)$	0	2.5	10	11.25	22.5	18.75	65
$x^2p(x)$	0	50	400	1012.5	2700	5625	9787.5

From the third row, $\mu = 65$. From the last row,

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 9787.5 - 65^2 = 5562.5, \qquad \sigma = \sqrt{5562.5} \div 74.58.$$

The mean is $\$65$ and the standard deviation is about $\$74.58$.

Your turn. Let $\$x$ be the amount the thief takes. Each friend is chosen with probability $\frac{1}{10}$.

x	0	20	30	50	Sum
$p(x)$	0.4	0.1	0.3	0.2	1
$xp(x)$	0	2	9	10	21
$x^2p(x)$	0	40	270	500	810

From the third row, $\mu = 21$. From the last row,

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 810 - 21^2 = 369, \qquad \sigma = \sqrt{369} \div 19.21.$$

The mean is $\$21$ and the standard deviation is about $\$19.21$.

Textbook Homework

1

a

x	1	2	3	4	Sum
$p(x)$	0.3	0.5	0.1	0.1	1
$xp(x)$	0.3	1	0.3	0.4	2
$(x-\mu)^2$	1	0	1	4	—
$(x-\mu)^2p(x)$	0.3	0	0.1	0.4	0.8

$\mu = 2$, $\mathrm{Var}(X) = 0.8$

b $\sigma = \sqrt{0.8} \div 0.89$

2

a

x	1	2	3	4	Sum
$p(x)$	0.3	0.5	0.1	0.1	1
$xp(x)$	0.3	1	0.3	0.4	2
x^2	1	4	9	16	—
$x^2p(x)$	0.3	2	0.9	1.6	4.8

b $\mathrm{Var}(X) = 4.8 - 2^2 = 0.8$

3

a $\mathrm{E}(X) = 2$, $\mathrm{E}(X^2) = 6$, $\mathrm{Var}(X) = 6 - 2^2 = 2$

b $\mathrm{E}(X) = 3$, $\mathrm{E}(X^2) = 10$, $\mathrm{Var}(X) = 10 - 3^2 = 1$

c $\mathrm{E}(X) = 0$, $\mathrm{E}(X^2) = 2.6$, $\mathrm{Var}(X) = 2.6 - 0^2 = 2.6$

d $\mathrm{E}(X) = 2.8$, $\mathrm{E}(X^2) = 9.2$, $\mathrm{Var}(X) = 9.2 - 2.8^2 = 1.36$

In each part the definition gives the same variance.

4 **a** (i) $\mathrm{E}(Y) = 2$, $\mathrm{Var}(Y) = 1$, $\sigma = 1$
(ii) $\mathrm{E}(Z) = 2$, $\mathrm{Var}(Z) = 4$, $\sigma = 2$
(iii) $\mathrm{E}(V) = 1$, $\mathrm{Var}(V) = 0.8$, $\sigma \div 0.89$
(iv) $\mathrm{E}(W) = 3$, $\mathrm{Var}(W) = 0.8$, $\sigma \div 0.89$
b (i) Both sets of data are centred around 2 and the expected value of each is unsurprisingly 2. The second data set is more spread out. In fact, in moving from Y to Z the distances from the mean to each data point have been doubled and the standard deviation is doubled.
(ii) The data has been 'flipped over', but is no more spread out than before: the variance is unchanged. You may notice that $W = 4 - V$.

5 $\mathrm{E}(X) = 2$, $\mathrm{Var}(X) = (2-2)^2 \times 1 = 0$

6 **a** $\mathrm{E}(J) = 1.55$, $\mathrm{E}(J^2) = 4.45$, $\mathrm{Var}(J) = 4.45 - 1.55^2 = 2.0475 \div 2.05$
$\mathrm{E}(L) = 1.4$, $\mathrm{E}(L^2) = 2.8$, $\mathrm{Var}(L) = 2.8 - 1.4^2 = 0.84$
b Over the season John might be expected to score more baskets, because his expected value is higher.
c Liam is the more consistent player, with the lower variance. Coaches may prefer a more consistent player, particularly if it is more important to score *some* goals, rather than the maximum number. This may also be a sign that John needs to work on the consistency of his game.

7 **a** Each outcome has probability $\frac{1}{3}$. This is a uniform distribution.
b $\mathrm{E}(X) = 2$ **c** $\mathrm{Var}(X) = \frac{2}{3}$

8 **a** $\frac{55-65}{5} = -2$: two standard deviations below the mean.
b $\frac{54-72}{12} = -1.5$: one and a half standard deviations below the mean.
c The English score was more standard deviations below the mean than the Mathematics result, so it may be considered less impressive.

9 **a** Visual Arts is 1 standard deviation below the mean, Music is 1.75 standard deviations below the mean, hence the Visual Arts score is better.
b Earth Science is 2 standard deviations above the mean, Biology is 1.5 standard deviations above the mean, hence the Earth Science score is more impressive.
c Chinese is 2 standard deviations above the mean, Sanskrit is also 2 standard deviations above the mean, hence the scores are equally impressive.

10 **a** $\mathrm{E}(X) = 3.3$, $\sigma \div 1.45$
b 8 appears to be a long way from 3.3 and well removed from the rest of the data.
c 8 is 3.2 standard deviations above the mean and thus would be an outlier by this definition.
d $\mathrm{E}(X) = 3.15$, $\sigma \div 1.06$
e The mean and standard deviation have changed significantly, especially the standard deviation.
f Outliers are interesting values in any distribution and should be a flag to investigate more closely. Were results recorded correctly? Was there an error in the experiment, for example, did Jasmine use a more powerful bow with greater range, or maybe she used a new set of arrows with better fletching? It may, however, be that Jasmine is inconsistent, occasionally getting much better results, but often getting fairly poor results. In this case the large standard deviation is warranted as a measure of this distribution. Over 20 trials, a probability of 0.05 only represents one set of 10 shots, so a larger set of results may give a better picture of her long-term accuracy and reduce the impact of one strong result amongst many other weaker scores.

11 $k = \frac{1}{10}$, $\mathrm{E}(X) = 3$, $\sigma = 1$

12 **a** $\frac{1}{n}$

b $\mathrm{E}(X) = \frac{1}{n} \times \frac{1}{2}n(n+1) = \dfrac{n+1}{2}$

c $\mathrm{E}(X^2) = \frac{1}{n} \times \frac{1}{6}n(n+1)(2n+1) = \frac{1}{6}(n+1)(2n+1)$, so

$\mathrm{Var}(X) = \frac{1}{6}(n+1)(2n+1) - \frac{1}{4}(n+1)^2 = \frac{1}{12}(n^2-1)$

d With $n = 3$, the formulas give $\mathrm{E}(X) = 2$ and $\mathrm{Var}(X) = \frac{8}{12} = \frac{2}{3}$, as in Question 7. With $n = 6$, they give $\mathrm{E}(X) = 3\frac{1}{2}$ and $\mathrm{Var}(X) = \frac{35}{12} = 2\frac{11}{12}$, as for the die in Worked examples 5 and 6.

13 **a** Because $Z = X + a$,

$$\begin{aligned}\mathrm{E}(Z) &= \sum zP(Z=z)\\ &= \sum (x+a)P(X+a=x+a)\\ &= \sum (x+a)P(X=x)\\ &= \sum xP(X=x) + \sum aP(X=x)\\ &= \sum xP(X=x) + a\sum P(X=x)\\ &= \mu + a,\end{aligned}$$

because $\sum P(X=x) = 1$.

b Because $Z = kX$,

$$\begin{aligned}\mathrm{E}(Z) &= \sum zP(Z=z)\\ &= \sum (kx)P(kX=kx)\\ &= \sum (kx)P(X=x)\\ &= k \times \sum xP(X=x)\\ &= k\mu.\end{aligned}$$

14 **a** The mean of Z is $\mu + a$, by the previous question. Hence

$$\begin{aligned}\mathrm{Var}(Z) &= \mathrm{E}\big((Z-(\mu+a))^2\big)\\ &= \mathrm{E}\big((Z-a-\mu)^2\big)\\ &= \mathrm{E}\big((X-\mu)^2\big)\\ &= \mathrm{Var}(X)\end{aligned}$$

Hence the standard deviation of the new distribution remains σ. This is to be expected, because the distribution is no more spread out than previously.

b The mean of Z is $k\mu$, by the previous question. Hence

$$\begin{aligned}\mathrm{Var}(Z) &= \mathrm{E}\big((Z-k\mu)^2\big)\\ &= \mathrm{E}\big((kX-k\mu)^2\big)\\ &= k^2 \times \mathrm{E}\big((X-\mu)^2\big)\\ &= k^2\,\mathrm{Var}(X)\end{aligned}$$

Hence the standard deviation of the new distribution is $\sqrt{k^2\sigma^2} = k\sigma$.

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13C | Variance and standard deviation

Past-Paper Homework - Worked Solutions

PP 1 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q1**

Let $X = 1$ for a success and $X = 0$ for a failure. Then $p(1) = 0.3$ and $p(0) = 0.7$.

x	0	1	Sum
$p(x)$	0.7	0.3	1
$xp(x)$	0	0.3	0.3
$x^2p(x)$	0	0.3	0.3

So $\mu = 0.3$ and $\mathrm{E}(X^2) = 0.3$, and

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 0.3 - 0.3^2 = 0.3 - 0.09 = 0.21.$$

Answer (D). Option (A), 0.09, is μ^2 only. Option (B), 0.7, is the probability of failure, and option (C), 0.3, is the mean.

PP 2 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)(iii)**

(iii) Put $a = 0.4116$ and $b = 0.2646$ into the table, and add rows for $xp(x)$ and $x^2p(x)$.

x	0	1	2	3	4	Sum
$p(x)$	0.2401	0.4116	0.2646	0.0756	0.0081	1
$xp(x)$	0	0.4116	0.5292	0.2268	0.0324	1.2
$x^2p(x)$	0	0.4116	1.0584	0.6804	0.1296	2.28

From the third row, $\mu = 1.2$. From the last row, $\mathrm{E}(X^2) = 2.28$. Hence

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = 2.28 - 1.2^2 = 2.28 - 1.44 = 0.84.$$

PP 3 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q8**

The Bernoulli variable takes the value 1 with probability p and the value 0 with probability $1 - p$. So

$$\mathrm{E}(X) = 0 \times (1-p) + 1 \times p = p, \qquad \mathrm{E}(X^2) = 0^2 \times (1-p) + 1^2 \times p = p,$$
$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = p - p^2 = p(1-p).$$

The condition $\mathrm{E}(X) = 5\,\mathrm{Var}(X)$ becomes $p = 5p(1-p)$. Since $p \neq 0$, divide both sides by p:

$$1 = 5(1-p), \qquad 1 - p = \tfrac{1}{5}, \qquad p = \tfrac{4}{5}.$$

(Check: $\mathrm{E}(X) = \frac{4}{5}$ and $\mathrm{Var}(X) = \frac{4}{5} \times \frac{1}{5} = \frac{4}{25}$, and $5 \times \frac{4}{25} = \frac{4}{5}$.)

Answer (B). Option (C), $\frac{1}{5}$, is the value of $1 - p$, not p. For options (A) and (D), $\frac{1}{2}$ gives $\mathrm{E}(X) = \frac{1}{2}$ but $5\,\mathrm{Var}(X) = \frac{5}{4}$, and $\frac{3}{5}$ gives $\mathrm{E}(X) = \frac{3}{5}$ but $5\,\mathrm{Var}(X) = \frac{6}{5}$.

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13D | Sampling

Lesson Examples

C1 **Tutor demo. a** There are 8 equally likely outcomes, so:

x	0	1	2	3	Sum
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1
$xp(x)$	0	$\frac{3}{8}$	$\frac{6}{8}$	$\frac{3}{8}$	$1\frac{1}{2}$
$x^2p(x)$	0	$\frac{3}{8}$	$\frac{12}{8}$	$\frac{9}{8}$	3

$\mu = 1\frac{1}{2},\ \ \sigma^2 = 3 - \left(1\frac{1}{2}\right)^2 = \frac{3}{4},\ \ \sigma = \sqrt{0.75} \doteqdot 0.87$

b Divide each frequency by 100:

x	0	1	2	3	Sum
f	14	35	39	12	100
f_r	0.14	0.35	0.39	0.12	1
xf_r	0	0.35	0.78	0.36	1.49
x^2f_r	0	0.35	1.56	1.08	2.99

$\bar{x} = 1.49,\ \ s^2 = 2.99 - 1.49^2 = 0.7699 \doteqdot 0.77,\ \ s = \sqrt{0.7699} \doteqdot 0.88$

c The sample mean 1.49 is very close to $\mu = 1.5$, and $s \doteqdot 0.88$ is very close to $\sigma \doteqdot 0.87$. The sample results are consistent with the theoretical results.

Your turn. a There are 4 equally likely outcomes, so:

x	0	1	2	Sum
$p(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$	1
$xp(x)$	0	$\frac{2}{4}$	$\frac{2}{4}$	1
$x^2p(x)$	0	$\frac{2}{4}$	$\frac{4}{4}$	$1\frac{1}{2}$

$\mu = 1,\ \ \sigma^2 = 1\frac{1}{2} - 1^2 = \frac{1}{2},\ \ \sigma = \sqrt{0.5} \doteqdot 0.71$

b Divide each frequency by 50:

x	0	1	2	Sum
f	11	27	12	50
f_r	0.22	0.54	0.24	1
xf_r	0	0.54	0.48	1.02
x^2f_r	0	0.54	0.96	1.50

$\bar{x} = 1.02,\ \ s^2 = 1.50 - 1.02^2 = 0.4596 \doteqdot 0.46,\ \ s = \sqrt{0.4596} \doteqdot 0.68$

c The sample mean 1.02 is close to $\mu = 1$, and $s \doteqdot 0.68$ is a little below $\sigma \doteqdot 0.71$. The sample results are consistent with the theoretical results.

C2 **Tutor demo. a** It is a survey, because only 50 of the 800 students are asked.

b Divide each frequency by 50:

x	0	1	2	3	4	5	Sum
f	8	17	15	6	3	1	50
f_r	0.16	0.34	0.30	0.12	0.06	0.02	1
xf_r	0	0.34	0.60	0.36	0.24	0.10	1.64
x^2f_r	0	0.34	1.20	1.08	0.96	0.50	4.08

$\bar{x} = 1.64$, $s^2 = 4.08 - 1.64^2 = 1.3904$, $s = \sqrt{1.3904} \div 1.18$

c Only some of the students are asked, and a different random choice of 50 students would give different results. So the sample mean will seldom be exactly the population mean μ.

Your turn. a It is a survey, because only some people are asked.

b Divide each frequency by 40:

x	0	1	2	3	4	Sum
f	10	14	9	5	2	40
f_r	0.25	0.35	0.225	0.125	0.05	1
xf_r	0	0.35	0.45	0.375	0.2	1.375
x^2f_r	0	0.35	0.9	1.125	0.8	3.175

$\bar{x} = 1.375$, $s^2 = 3.175 - 1.375^2 = 1.284375$, $s = \sqrt{1.284375} \div 1.13$

c No. An internet survey includes only people visiting that website, and only those who choose to answer, so the sample is in no way random.

C3 **Tutor demo. a** Divide each frequency by 40:

x	0	1	2	3	4	5	Sum
f	6	12	11	7	3	1	40
f_r	0.15	0.3	0.275	0.175	0.075	0.025	1
xf_r	0	0.3	0.55	0.525	0.3	0.125	1.8
x^2f_r	0	0.3	1.1	1.575	1.2	0.625	4.8

$\bar{x} = 1.8$ goals, $s^2 = 4.8 - 1.8^2 = 1.56$, $s = \sqrt{1.56} \div 1.25$ goals

b For example: Has the team changed its players or its coach? Will it play the same opponents next season? Were there injuries, or unusual weather, last season? Is one season of 40 matches enough data?

Your turn. a Divide each frequency by 25:

x	2	3	4	5	6	Sum
f	3	6	9	5	2	25
f_r	0.12	0.24	0.36	0.2	0.08	1
xf_r	0.24	0.72	1.44	1	0.48	3.88
x^2f_r	0.48	2.16	5.76	5	2.88	16.28

$\bar{x} = 3.88$ eggs, $s^2 = 16.28 - 3.88^2 = 1.2256$, $s = \sqrt{1.2256} \div 1.11$ eggs

b No. There are no equally likely possible outcomes here, and no known population. The relative frequencies are only estimates from 25 nests, and another 25 nests would give different values.

C4 **Tutor demo.** The correction factor is $\frac{40}{39}$.

$$\textbf{corrected variance} = 1.56 \times \tfrac{40}{39} = 1.6, \qquad \textbf{corrected standard deviation} = \sqrt{1.6} \div 1.26.$$

These are slightly larger than $s^2 = 1.56$ and $s = \sqrt{1.56} \div 1.25$.

Your turn. The correction factor is $\frac{16}{15}$.

$$\textbf{corrected variance} = 3.75 \times \tfrac{16}{15} = 4, \qquad \textbf{corrected standard deviation} = \sqrt{4} = 2.$$

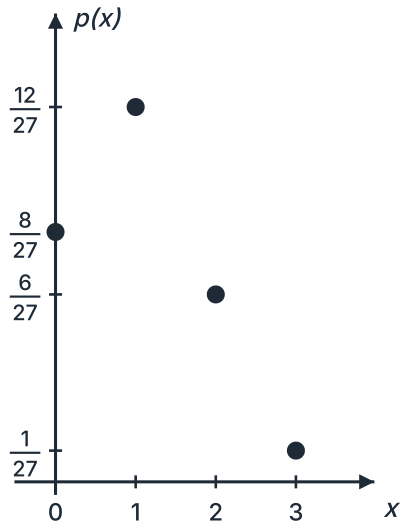
These are larger than $s^2 = 3.75$ and $s = \sqrt{3.75} \div 1.94$. The sample is small, so the correction makes more difference.

Textbook Homework

1 **a**

x	0	1	2	3	Sum
$p(x)$	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$	1
$xp(x)$	0	$\frac{12}{27}$	$\frac{12}{27}$	$\frac{3}{27}$	1
$x^2p(x)$	0	$\frac{12}{27}$	$\frac{24}{27}$	$\frac{9}{27}$	$1\frac{2}{3}$

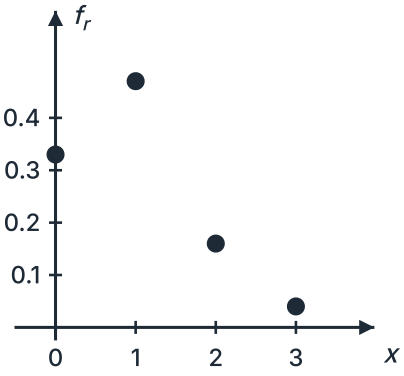
$\mu = 1, \ \sigma^2 = 1\frac{2}{3} - 1^2 = \frac{2}{3}, \ \sigma \div 0.82$



b

x	0	1	2	3	Sum
f	33	47	16	4	100
f_r	0.33	0.47	0.16	0.04	1
xf_r	0	0.47	0.32	0.12	0.91
x^2f_r	0	0.47	0.64	0.36	1.47

$\bar{x} = 0.91, \ s^2 = 1.47 - (0.91)^2 = 0.6419, \ s \div 0.80$



c The sample results are a little below what is predicted by the theoretical probabilities.

2 **a** $\mu = 7, \ \sigma^2 = \frac{210}{36} = \frac{35}{6}, \ \sigma \div 2.42$

4 **c** (iii)

x	0	1	2	3	4	5	Sum
$p(x)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1
$xp(x)$	0	$\frac{10}{36}$	$\frac{16}{36}$	$\frac{18}{36}$	$\frac{16}{36}$	$\frac{10}{36}$	$\frac{70}{36}$
$x^2p(x)$	0	$\frac{10}{36}$	$\frac{32}{36}$	$\frac{54}{36}$	$\frac{64}{36}$	$\frac{50}{36}$	$\frac{210}{36}$

$\mu \div 1.94, \ \sigma^2 = \frac{210}{36} - \left(\frac{70}{36}\right)^2, \ \sigma \div 1.43$

7 **a** $\mu = \sum xp(x) = 3.441, \ \mathrm{E}(X^2) = \sum x^2p(x) = 17.901, \ \sigma^2 = 17.901 - 3.441^2 \div 6.06, \ \sigma \div 2.46$

12 **a** Later people taking part in the experiment will be influenced by earlier guesses, particularly if the previous guesses have been measured for accuracy. Perhaps students could record their estimate, or draw their estimated shape, at the same time and before any measuring occurs. Perhaps students go into a separate room for the experiment.

14 **a** $m-k$ is the number of serial numbers not yet discovered in the range from 1 to m. If these serial numbers are spread between the k gaps, the average size of the gap (number of undiscovered serials) is $\frac{m-k}{k}$.

b The gap of $\frac{m-k}{k}$ integers should extend past m to $m+\frac{m-k}{k}$. Using this estimate the last serial will be:

$$N=m+\frac{m-k}{k}=m+\frac{m}{k}-1$$

Chapter 13 Discrete Distributions - Homework Answers

Lesson 13D | Sampling

Past-Paper Homework - Worked Solutions

> **PP 1** **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q2**
>
> The proportion $\hat{p}$ is the relative frequency of the outcome 3 in the 18 rolls, that is, the number of rolls with an outcome of 3 divided by 18. So exactly 12 rolls with an outcome of 3 corresponds to $\hat{p} = \frac{12}{18} = \frac{2}{3}$, and 'at most 12' (12 or fewer) means $\hat{p} \leq \frac{2}{3}$. The probability is $P\left(\hat{p} \leq \frac{2}{3}\right)$.
>
> **Answer B.** Option D is the probability of *at least* 12 rolls with an outcome of 3. Options A and C use $\frac{1}{6}$, which is the probability of a 3 on one roll, not the proportion $\frac{12}{18}$.

Chapter 13 Discrete Distributions - Homework Answers

Chapter 13 Review | Chapter review exercise

1 **a** Numeric, continuous **b** Numeric, discrete **c** Numeric, discrete (and infinite) **d** Categorical

2 **a** Yes **b** No (the probabilities add to 1.2) **c** No (the probabilities add to $\frac{19}{24}$)

3 The probabilities are not all positive, do not sum to 1, and are not all less than 1.

4 **a** The $xp(x)$ row is 0, 0.3, 0.8, 0.3, so $\mathrm{E}(X) = 1.4$.
b The $xp(x)$ row is -1, -0.1, 0, 0.3, so $\mathrm{E}(X) = -0.8$.

5 **a** Let the cost of the meal be $\$X$.

$$\mathrm{E}(X) = \frac{27 \times 4 + 32 \times 2 + 23 \times 1 + 25 \times 2}{9} = \frac{245}{9} \div 27.22$$

His expected cost is about $\$27.22$.
b His expected cost is $52 \times \frac{245}{9} \div 1415.56$, that is, about $\$1415.56$.

6 **a** $\mathrm{E}(X) = 2$, $\mathrm{Var}(X) = 1$, $\sigma = 1$ **b** $\mathrm{E}(X) = 5.1$, $\mathrm{Var}(X) = 0.69$, $\sigma \div 0.83$

7 **a** $\mathrm{E}(X) = 2$, $\mathrm{E}(X^2) = 5$, $\mathrm{Var}(X) = 1$ **b** $\mathrm{E}(X) = 5.1$, $\mathrm{E}(X^2) = 26.70$, $\mathrm{Var}(X) = 0.69$

8 **a** $\mathrm{E}(X) = 1.9$, $\mathrm{Var}(X) = 0.49$, $\sigma = 0.7$ **b** $\mathrm{E}(X) = 2$, $\mathrm{Var}(X) = 2.6$, $\sigma \div 1.61$

9 Expected value is a measure of central tendency: it measures the centre of the data set. It may also be thought of as a weighted mean (weighted by the probabilities of the distribution). If the experiment is carried out experimentally a large number of times we would expect that the average of the outcomes would approach the expected value.

10 The standard deviation is the square root of the variance. Both measure the spread of the data, so that a distribution with a larger standard deviation is more spread out than a distribution with a smaller standard deviation. Both are zero if the distribution only takes one value, that is, if it is not spread out at all. If the distribution is stretched (multiplied) by a constant k the standard deviation also increases by a factor k.

11 **a** 12, 8, $2\sqrt{2}$ **b** 11, 2, $\sqrt{2}$ **c** 17, 18, $3\sqrt{2}$

12 **a**

x	5	6	7	8	9
$p(x)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

$\mathrm{E}(X) = 7$, $\mathrm{Var}(X) = 2$, $\sigma = \sqrt{2}$.